

DATA3 NF270 — Choosing B(c), the transport regime, and whether experiments inform each other

Single-salt diafiltration: NaCl · CaCl₂ · LaCl₃ | per-response AIC · Peclet regime · multi-experiment pooling

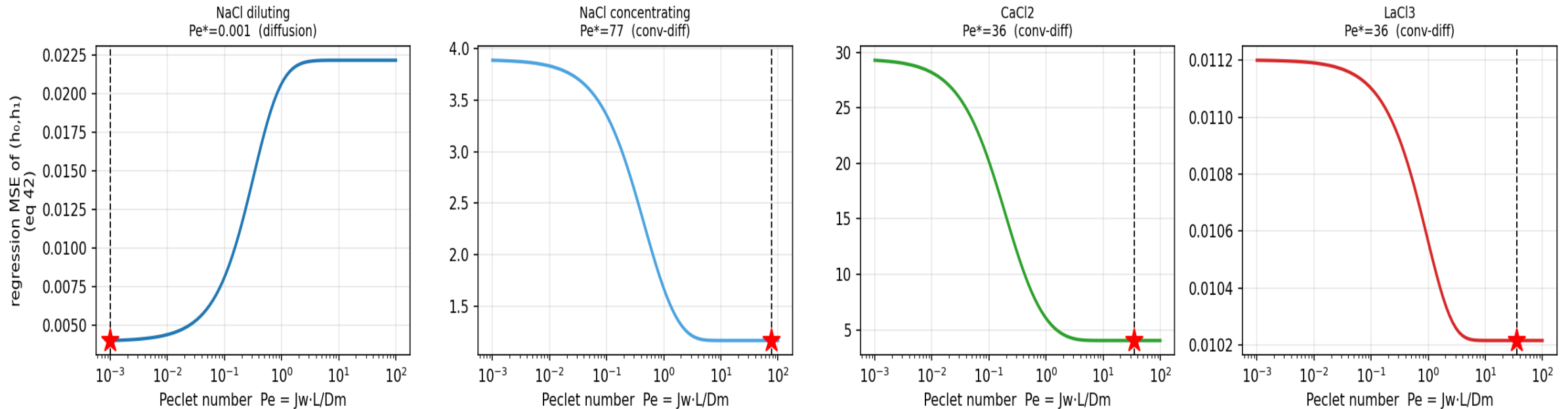
What sets the solute permeability B , and does it depend on concentration?

- B is the membrane solute permeability: $J_s = J_w \cdot B \cdot (c_{in} - c_H)$.
- DATA1/DATA2 transport theory: $B = D_m \cdot H / L$ — a partition (H) and a diffusion length.
- Whether B is constant or concentration-dependent is set by the Peclet number $Pe = J_w \cdot L / D_m$:
 - $Pe \rightarrow 0$ (diffusion-dominated): $B = D_m H / L = \text{CONSTANT}$
 - $Pe \sim 1+$ (convection + diffusion): $B = J_w \cdot \sum \beta_i c^i$ (concentration-dependent Taylor form)
- Under a constant- B fit, σ and B and L_p are coupled and σ rails — the identifiability problem we solve here.

→ Three threads: which $B(c)$ form (AIC), why it depends on c (Peclet), and can pooling experiments fix identifiability.

Regression MSE vs Pe ($\triangleq$ DATA2 Fig 8): which regime each salt is in

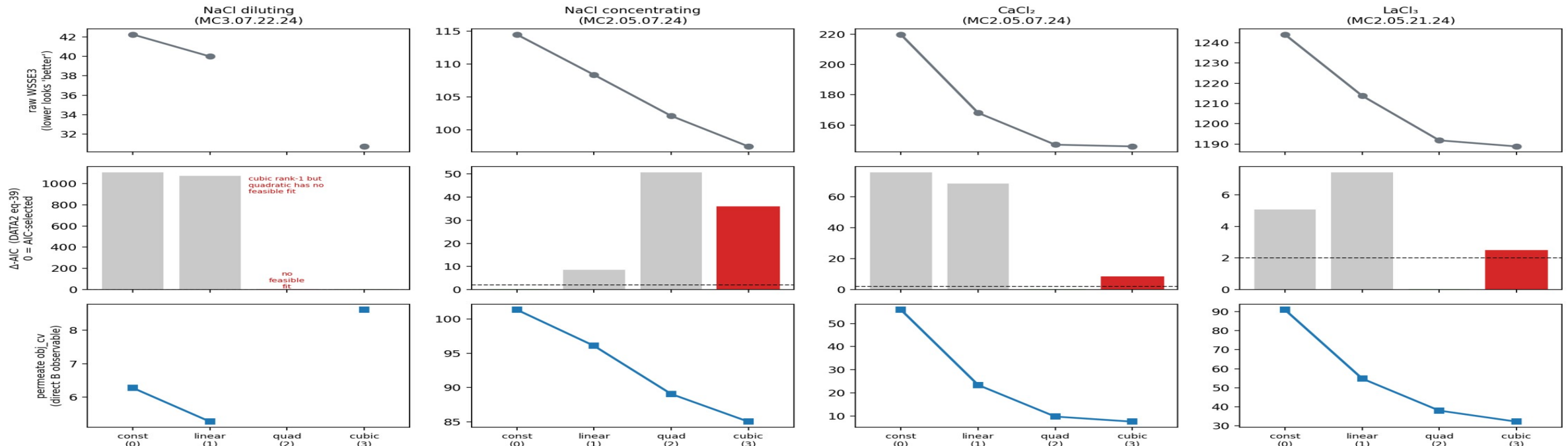
DATA3 Peclet analysis ($\triangleq$ DATA2 Fig 8A): regression MSE vs Pe — a minimum at $Pe \geq 1$ means convection+diffusion, i.e. B is concentration-dependent (not the $Pe \rightarrow 0$ constant- B limit)



→ Diluting NaCl → $Pe \rightarrow 0$ (diffusion, $B \approx \text{const}$). CaCl₂ / concentrating-NaCl / LaCl₃ → $Pe \gg 1$ (convection+diffusion, $B(c)$). The regime direction is the robust signal.

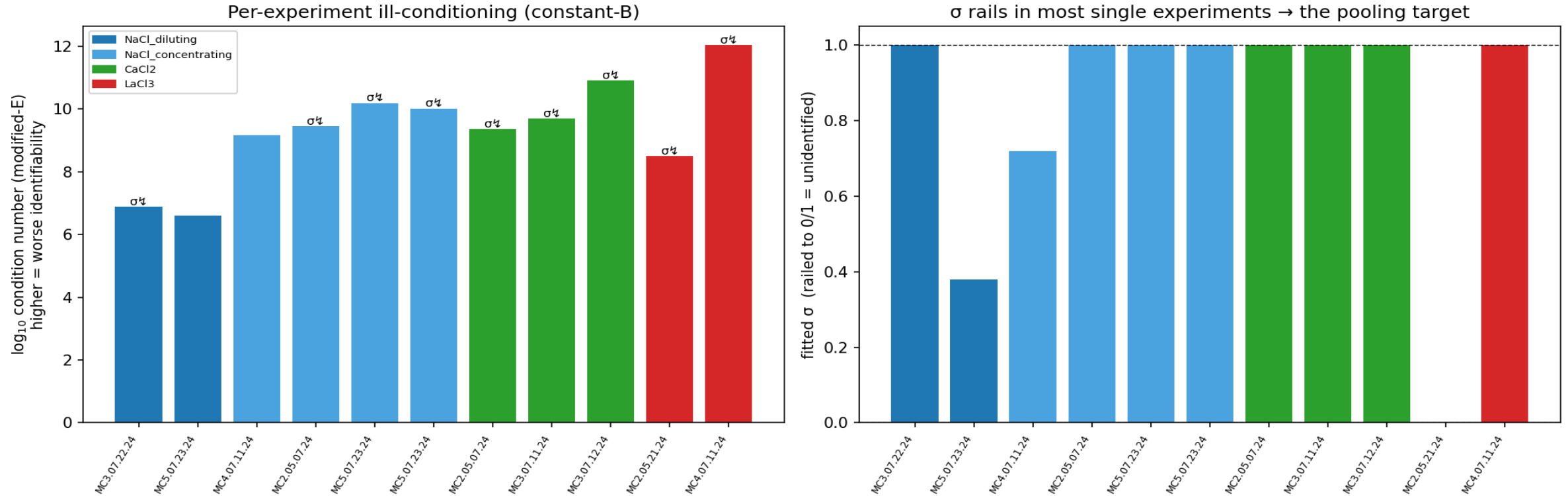
Order chosen by the per-channel AIC (eq 39), not raw WSSE3

DATA3 B(c) order selection — DATA2 per-response AIC, not raw WSSE3. Top: raw WSSE3 always falls with order (the overfit trap). Middle: Δ -AIC selects low order (green) and rejects cubic (red) wherever a full candidate set converged; hatched = no feasible fit (cubic 'wins' on the two diluting NaCl sheets only because quadratic is missing). Bottom: the permeate channel (the direct B observable) collapses for CaCl_2 = real low-order B(c); it barely moves for NaCl = B(c) weak.



→ CaCl_2 → quadratic/linear, cubic REJECTED ($\Delta=8-25$). NaCl cubic 'wins' are mass-channel/convergence artifacts. Use the per-response AIC, not the pooled JSON AIC.

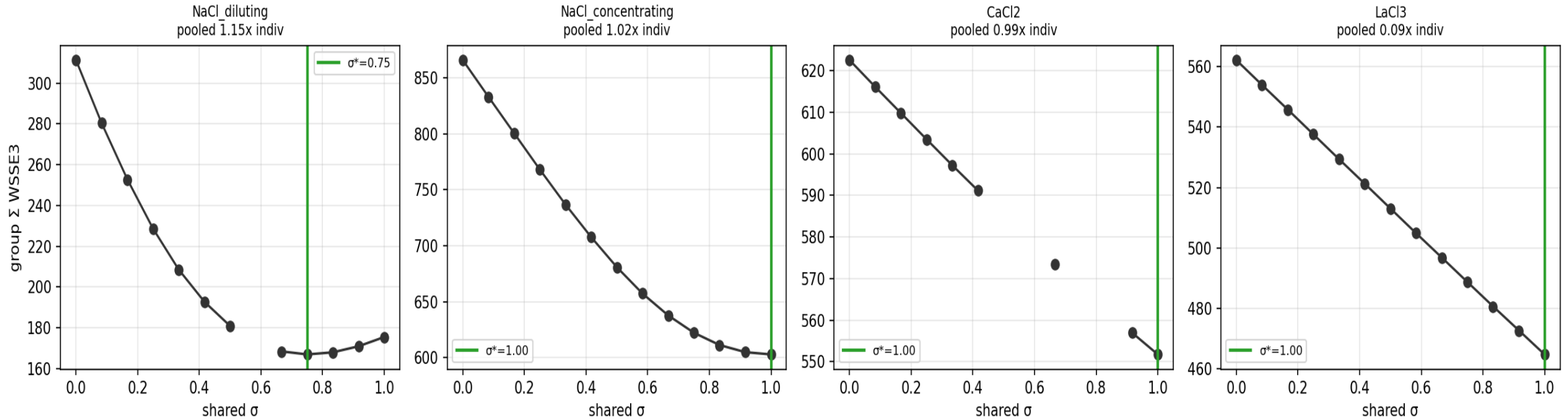
Per-experiment σ rails; the σ -B- L_p landscape is ill-conditioned



$\rightarrow$ σ rails in 9/11 single experiments; condition numbers 10^6 – 10^{12} . A single experiment cannot pin σ .

One shared σ per salt-regime group costs ~nothing — and comes off the rail

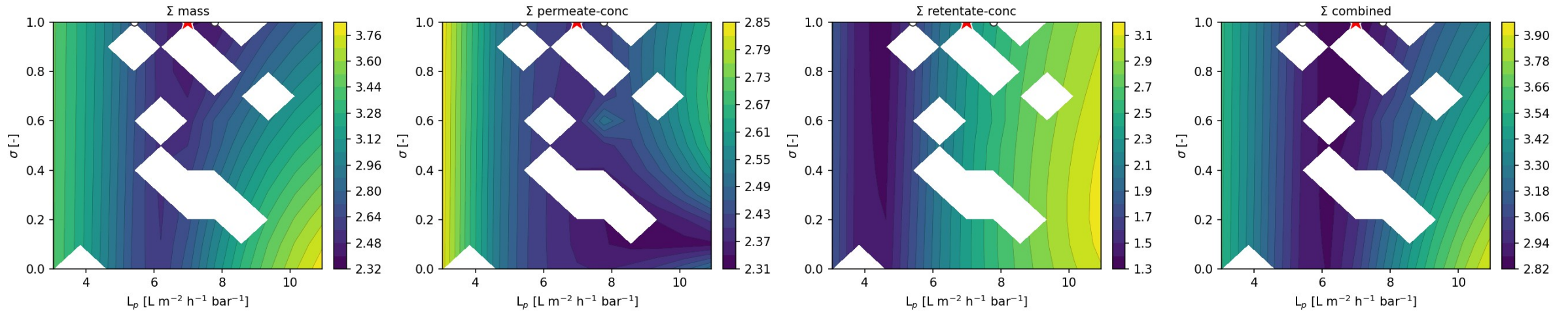
Stage 3 — shared- σ profile per group: a sharp interior minimum = pooling identifies σ (resolves the single-experiment rail)



→ Cost $\approx 1\times$ (sharing σ is nearly free → σ is a salt property). Diluting-NaCl identifies $\sigma^* = 0.75$, OFF the individual rails (1.0/0.38).

Summing the objective surfaces = summing Fisher information → a sharper pooled minimum

STACKED $L_p \times \sigma$ — CaCl₂ (single): 3 experiments summed (other params profiled). ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint-fit seed. stacked min ★ $L_p=6.99$, $\sigma=1.00$



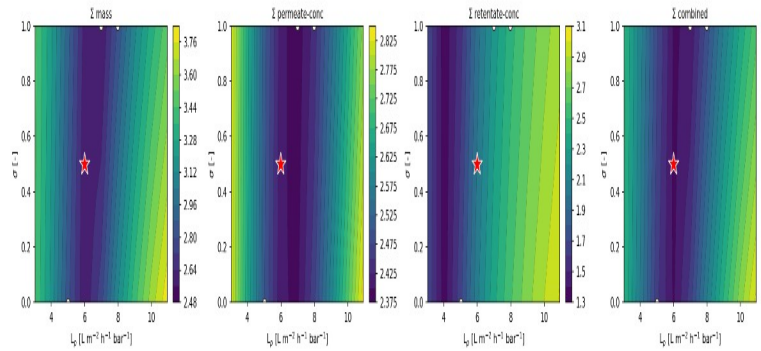
Per channel across a group: Σ mass → L_p , Σ permeate-conc → B , Σ retentate-conc → σ . White ○ = each experiment's scattered own minimum; red ★ = the pooled minimum (joint-fit seed). The transport model is evaluated at every node (slice = forward-sim, legacy calc_contour_2d; profile = re-solve $B(c)$).

→ Where individuals are sloppy, the summed surface localizes — the visual proof that pooling improves identifiability.

Each concentration-function of B reshapes the σ/L_p identifiability (CaCl_2)

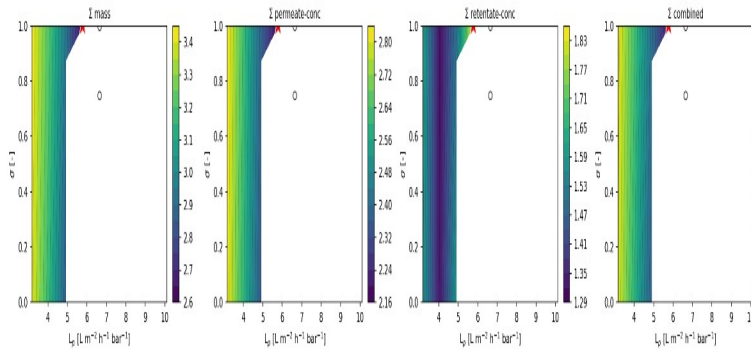
constant

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=slice - slice (B(c) held at fit, forward-sim): 3 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=0.00$, $\sigma=0.50$



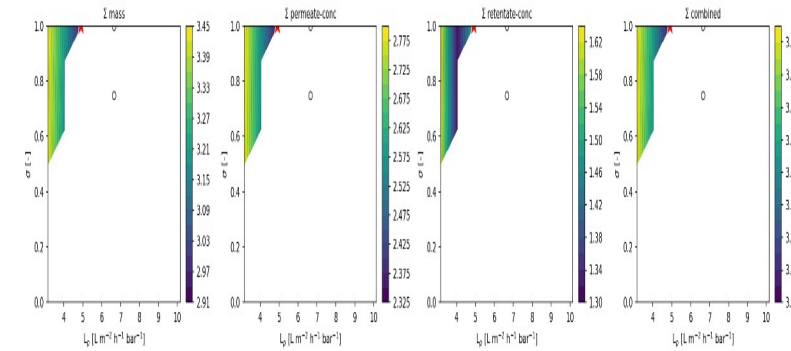
linear

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=pol2 - slice (B(c) held at fit, forward-sim): 3 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=3.78$, $\sigma=1.00$



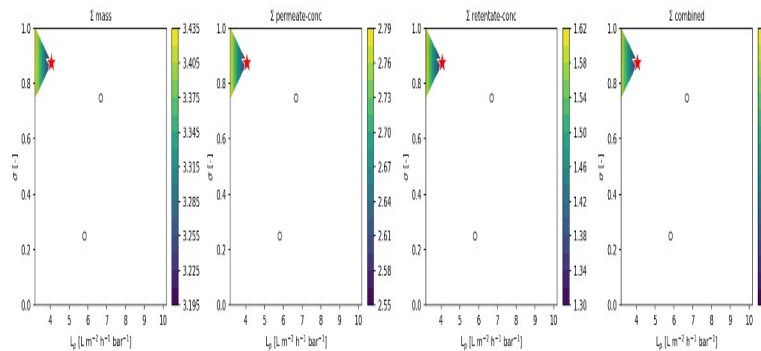
quadratic

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=pol2 - slice (B(c) held at fit, forward-sim): 3 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=4.91$, $\sigma=1.00$



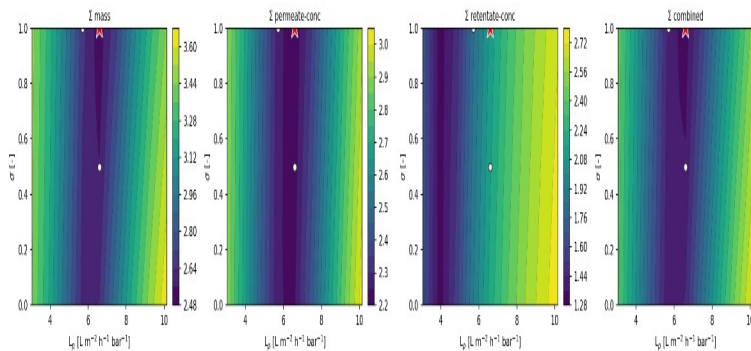
cubic

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=pol3 - slice (B(c) held at fit, forward-sim): 3 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=4.06$, $\sigma=0.88$



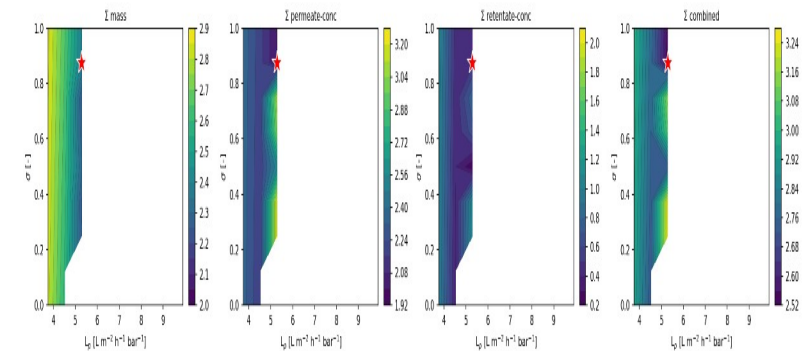
saturating

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=sat - slice (B(c) held at fit, forward-sim): 3 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=6.60$, $\sigma=1.00$



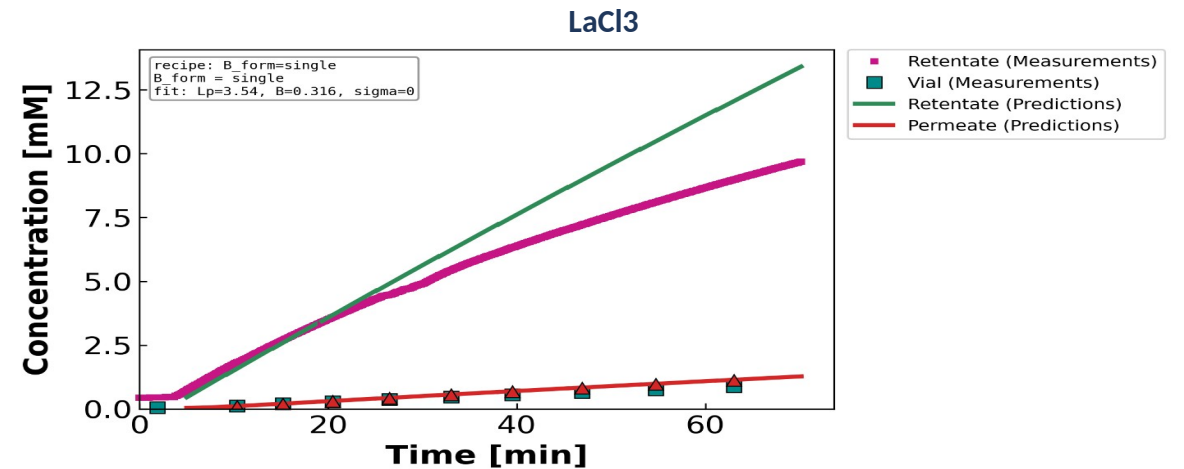
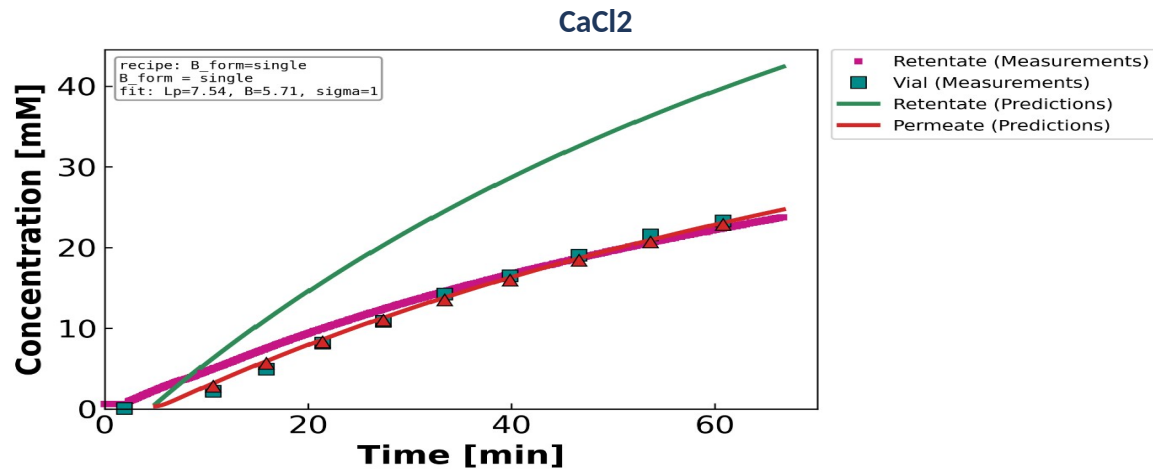
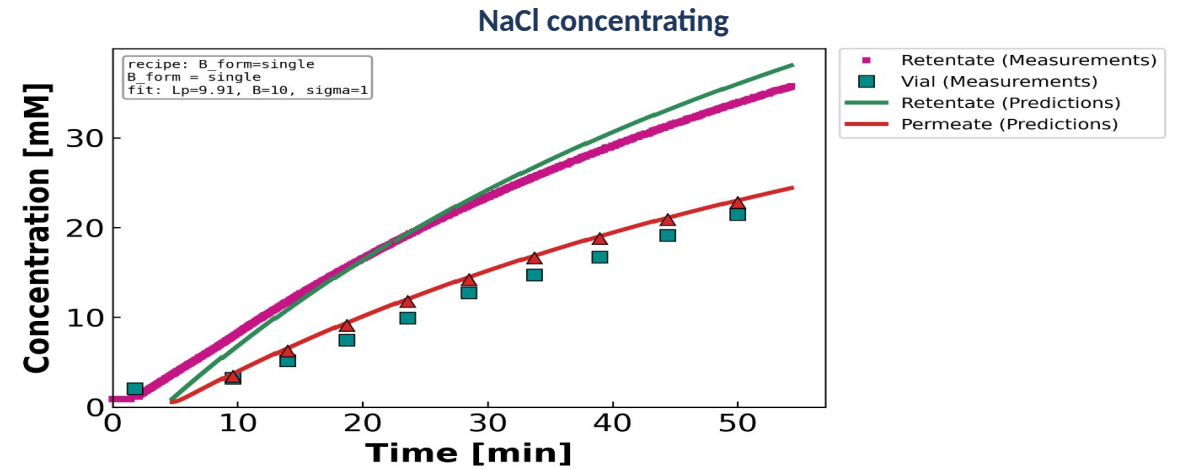
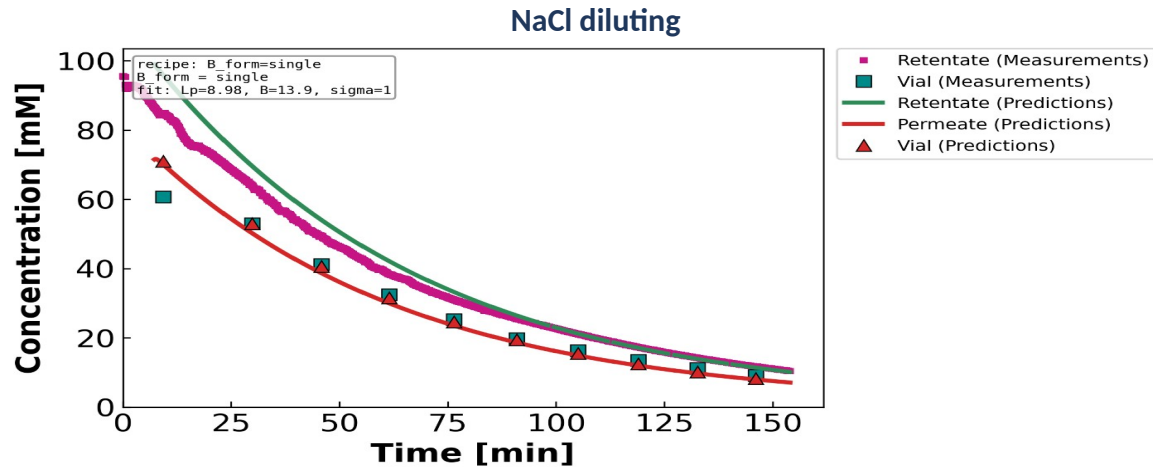
Donnan

STACKED $L_p \times \sigma$ — CaCl₂ - B-form=donnan - slice (B(c) held at fit, forward-sim): 1 experiments summed. ○ = each experiment's own min (scattered = sloppy); ★ = pooled min / joint fit seed. stacked min ★ $L_p=5.28$, $\sigma=0.88$



→ Comparing forms: constant rails σ ; the concentration-dependent forms relax the σ/L_p trade-off differently.

The first B-form (constant B) at the optimized parameters



→ Constant-B tracks NaCl well (flat B); for CaCl₂/LaCl₃ the constant-B residual motivates B(c).

1. B is constant when diffusion dominates ($Pe \rightarrow 0$) and concentration-dependent when convection matters ($Pe \gg 1$). The Peclet analysis separates the salts: diluting-NaCl diffusion-limited; $\text{CaCl}_2/\text{LaCl}_3/\text{conc-NaCl}$ convection.
2. Among concentration-functions of B, the DATA2 per-response AIC selects LOW order — quadratic/linear for CaCl_2 , cubic rejected. Raw WSSE3 over-selects; the per-channel likelihood is the honest criterion.
3. A single experiment cannot identify σ (it rails; cond $10^6 - 10^{12}$). Pooling experiments of the same salt+regime shares σ at ~zero cost and resolves the rail (diluting-NaCl $\sigma^* = 0.75$). Stacked objective landscapes show the experiments informing each other: the summed surface has a sharper minimum than any individual.
4. Same NF270 membrane across DATA3 $\rightarrow L_p$ is consistent; the within-group scatter was identifiability noise that pooling removes. The stacked minimum is the warm-start for the joint multi-experiment fit.